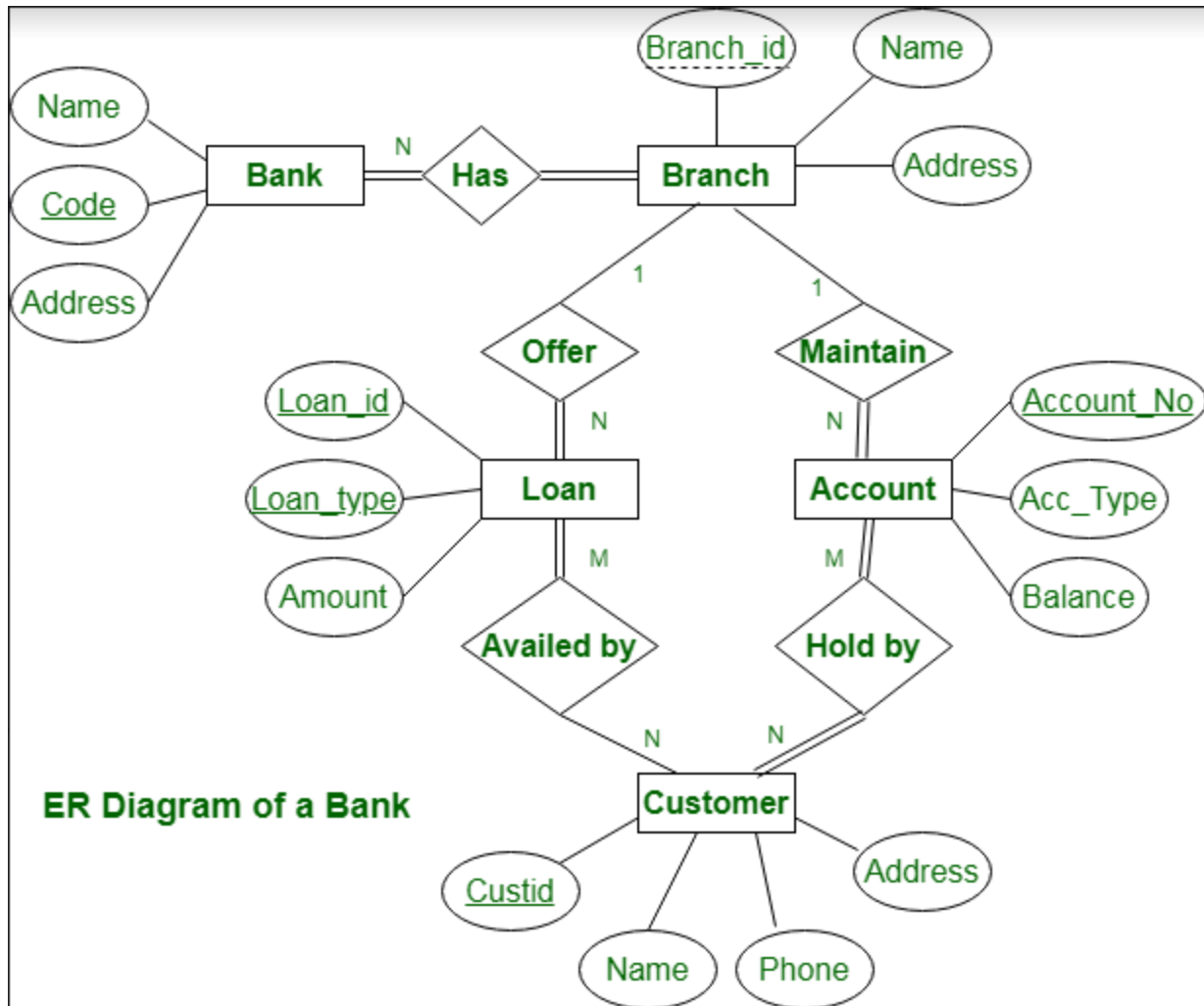




Features to Implement

Based on the ER diagram, the project needs to manage core banking operations. Here are the features:

- **Bank Management:**
 - **Bank:** Create, read, update, and delete (**CRUD**) operations for banks. A bank has a Name, Code, and Address.
 - **Branch:** CRUD operations for bank branches. A branch has a Branch_id, Name, and Address.
 - **Relationship:** A **Bank** Has many **Branches**.

- **Customer and Account Management:**
 - **Customer:** CRUD operations for customers. A customer has a unique **Custid**, **Name**, **Address**, and **Phone**.
 - **Account:** CRUD operations for accounts. An account has an **Account_No**, **Acc_Type**, and **Balance**.
 - **Relationship:** A **Branch** **Maintains** many **Accounts**, and a **Customer** **Holds** many **Accounts**. This is a many-to-many relationship between a customer and an account, which is also linked to a branch.
- **Loan Management:**
 - **Loan:** CRUD operations for loans. A loan has a **Loan_id**, **Loan_type**, and **Amount**.
 - **Relationship:** A **Branch** can **Offer** many **Loans**, and a **Customer** can be **Availed by** many **Loans**. This is a many-to-many relationship between a customer and a loan, linked to a branch.

Project Flow

The project flow would simulate a basic banking application. Since the ER diagram doesn't include specific user roles (like a bank teller or a customer), the flow below assumes a high-level system for administrative use.

1. **System Startup:** The application starts and initializes its connection to the database.
2. **Initial Display:** The user is presented with a **main menu** or dashboard. This menu should have options for:
 - Manage Banks
 - Manage Branches
 - Manage Customers
 - Manage Accounts
 - Manage Loans
 - Exit
3. **User Selection:** The user selects an option from the main menu. For example, they choose "**Manage Customers**".
4. **Customer Management Menu:** A new sub-menu is displayed with options for:
 - **Create a new Customer:** The system prompts for customer details (**Name**, **Address**, **Phone**). After entry, the new customer is saved to the database.
 - **View all Customers:** The system retrieves and displays a list of all existing customers.
 - **Update a Customer:** The user selects a customer and updates their details.
 - **Delete a Customer:** The user selects a customer to delete.
 - **Back to Main Menu:** The system returns to the main menu.

5. **Handling Other Entities:** The flow for managing **Banks**, **Branches**, **Accounts**, and **Loans** would follow a similar pattern, with dedicated menus for each entity.
6. **Relationship Handling:** When managing entities, the flow must account for the relationships. For example:
 - When creating a new **Account**, the user should be able to link it to an existing **Branch** and **Customer**.
 - When creating a new **Loan**, the user should link it to a specific **Branch** and **Customer**.
7. **Exit:** The user selects **Exit**, and the application closes, properly terminating the database connection.